Sitka

DESIGN SYSTEM REPORT

Gap Analysis & Implementation Strategy

A benchmark of Sitka against Shopify Polaris, Salesforce Lightning, Apple HIG, and shadcn/ui — with a phased roadmap to modern multi-platform parity.

VERSION	DATE	BENCHMARKED AGAINST
Sitka 1.1.0	May 2026	Polaris · Lightning · Apple HIG · shadcn

Contents

1. Executive Summary

Sitka 1.1.0 is a well-structured, aesthetically distinctive design system with strong multi-platform ambitions. It documents 33 components, 20+ patterns, and is deployed as a Next.js 15 documentation site with React and SwiftUI code samples.

Benchmarked against the four most influential design systems of 2026 — **Shopify Polaris**, **Salesforce Lightning DS 2**, **Apple Human Interface Guidelines**, and **shadcn/ui** — Sitka has meaningful strengths and material gaps across six dimensions:

DIMENSION	CURRENT STATE	GAP SEVERITY
Component breadth	33 components documented; ~8 exist in code but are undocumented	Medium
Token system	W3C DTCG format used but only primitives are in `tokens.json`; semantic tokens live only in CSS comments	High
Foundations	Strong visual coverage; missing layout grid, RTL, density, deep accessibility specs	High
Platform coverage	Web + iOS/macOS — no Android, no watchOS/visionOS	Medium
Patterns	Rich data patterns; missing wizard, activity feed, chat/AI conversation	Low–Medium
Tooling & governance	No Storybook, no npm package, no lifecycle taxonomy, Figma library disabled	High

> **Key Finding**
> The single highest-leverage action is fixing the P0 items: documenting 8 components that already exist in code, and moving all semantic tokens into `tokens.json`. These require no new design work — only documentation — yet they close the most visible gaps against every reference system.

2. Where Sitka Already Leads

Before enumerating gaps, it is worth noting the areas where Sitka is ahead of all four reference systems:

LIQUID GLASS AESTHETIC

- Formalized glass surface tokens with blur, specular, and dimensional shadow layers
- Spring-first motion system aligned to Apple's 2025 Liquid Glass language
- Specular highlights, sheet entry animations documented as tokens

AI AGENT STANDARDS

- Dedicated foundations page for AI UI patterns — no other reference DS has this
- Early-mover position on a category that will define the next decade of UI

MULTI-PLATFORM CODE SAMPLES

- Every component ships React, HTML, and SwiftUI samples
- iOS and macOS SwiftUI libraries documented
- W3C DTCG token format with Swift export

RICH PATTERNS LIBRARY

- Kanban, Gantt, Analytics Dashboard, Burn Trajectory, Team Capacity — production-grade patterns rare in any DS
- Collaboration, Media Player, Goal/Streak patterns go beyond table-stakes coverage

3. Component Gaps

3.1 Undocumented Components (Quick Wins)

These components already exist as `.tsx` files in `/src/components/ui/` but have no documentation page and are absent from navigation. Zero design work required — documentation only.

COMPONENT	FILE	EFFORT	IN SHADCN	IN POLARIS
Tabs	`Tabs.tsx`	Low	✓	✓
Select	`Select.tsx`	Low	✓	✓
Checkbox	`Checkbox.tsx`	Low	✓	✓
Radio	`Radio.tsx`	Low	✓	✓
Switch	`Switch.tsx`	Low	✓	—
Command Palette	`CommandPalette.tsx`	Low	✓	—
Toast	`Toast.tsx`	Low	✓	deprecated

> **Note on Form Controls**
> Checkbox, Radio, and Switch are currently buried inside the Form Controls page. All three reference systems document them as independent components with full props tables, ARIA specs, and code samples. They warrant standalone pages.

3.2 Missing Components — Universal (all 4 reference systems have these)

COMPONENT	WHY IT MATTERS	POLARIS	SHADCN	LIGHTNING	APPLE
Accordion / Collapsible	Universal disclosure pattern for FAQs, settings, nested content	✓	✓	✓	✓
Alert / Banner (persistent)	Page-level feedback — distinct from transient Snackbar/Toast	✓	✓	✓	✓
Pagination	Required by every table-driven screen	✓	✓	✓	✓
Textarea	Multi-line text input; missing from component docs entirely	✓	✓	✓	✓
Spinner	Inline/button-level loading indicator separate from Skeleton	✓	✓	✓	✓

3.3 Missing Components — High Priority (2–3 reference systems)

COMPONENT	CATEGORY	POLARIS	SHADCN	LIGHTNING	APPLE
File Upload / Drop Zone	Input	✓	—	✓	✓
Drawer (horizontal slide-in)	Overlay	—	✓	✓	✓
Navigation Menu (mega/tree)	Navigation	—	✓	✓	✓
Sidebar (documented component)	Navigation	—	✓	✓	✓
Toggle / Toggle Group	Input	—	✓	✓	✓
Menubar (desktop)	Navigation	—	✓	✓	✓
Divider / Separator	Layout	✓	✓	✓	—
Label (standalone)	Input	✓	✓	✓	—
Tag (multi-value chip)	Display	✓	—	✓	—
Hover Card	Overlay	—	✓	—	✓
Input OTP / PIN	Input	—	✓	—	✓
Resizable Panels	Layout	—	✓	✓	✓
Kbd (keyboard key display)	Display	✓	✓	—	—
Data Grid (advanced table)	Display	✓	✓	✓	—

4. Token System Gaps

Sitka's `tokens.json` uses the W3C DTCG format but currently only contains **color primitives**, **spacing**, and **typography**. All semantic tokens — the values actually consumed in components — exist only as CSS custom property conventions in `SITKA_CONTEXT.md` and are not formally published in the token file.

> **Critical Impact**
> The Token Export feature (`/tokens/export`) and the Swift token generation can only export what is in `tokens.json` . Until semantic tokens are added, the export pipeline is incomplete and Figma/Android sync cannot be fully automated.

4.1 Missing Token Categories

CATEGORY	EXAMPLES	PRESENT IN TOKENS.JSON
Semantic colors	`--background` , `--surface` , `--surface-raised` , `--border` , `--text-primary` , `--accent`	No
Motion / Duration	`--duration-fast: 150ms` , `--duration-base: 250ms` , `--duration-slow: 400ms`	No
Motion / Easing	`--ease-spring` , `--ease-out` , `--ease-in-out`	No
Shadow	`--shadow-card` , `--shadow-card-lifted` , `--shadow-sheet`	No
Border radius	`--radius-sm` , `--radius-md` , `--radius-lg` , `--radius-pill`	No
Z-index / Layer	`--z-dropdown: 100` , `--z-modal: 300` , `--z-toast: 400`	No
Opacity	`--opacity-disabled: 0.38` , `--opacity-overlay: 0.5`	No
Glass / Blur	`--glass-blur: 20px` , `--glass-saturation: 180%` , `--glass-bg`	No
Typography (display)	`--text-display: 24px bold` , `--text-micro: 10px` , `--text-nano: 9px`	Partial

4.2 Recommended Token File Structure

The W3C DTCG format supports composite types and aliases. The full token hierarchy should be:

LAYER	PURPOSE	EXAMPLE
Primitive	Raw values — color, size, time	`color.brand.500 = #34a865`
Semantic	Intent aliases referencing primitives	`color.surface = {color.neutral.50}`
Component	Component-scoped overrides	`button.primary.bg = {color.accent}`

Currently only the Primitive layer is populated. All three layers should be present for complete Figma sync and multi-platform export.

5. Foundation Gaps

5.1 Layout System

This is the most significant structural gap. Every reference system has a documented layout/grid system; Sitka has a Desktop Layout page but no formal column grid, no responsive breakpoints as tokens, and no layout primitive components.

WHAT'S MISSING	POLARIS EQUIVALENT	SHADCN EQUIVALENT
12-column responsive grid	Grid, Inline Grid	Tailwind grid utilities
Stack layout components	Block Stack, Inline Stack	—
Box (layout container)	Box, Bleed	—
Breakpoint tokens (sm/md/lg/xl)	Documented as px values	Tailwind default scale
Page layout template	Page, Layout	—

5.2 Accessibility Depth

Sitka has an Accessibility page and a Contrast page, but lacks the per-component accessibility specification that separates a documentation-level DS from a production-grade one.

- **ARIA roles per component** — What role, aria-label, aria-expanded, aria-controls does each component expose?
- **Keyboard navigation model** — Tab order, arrow key navigation in complex widgets (combobox, data grid, tree), focus trapping in modals
- **Focus management patterns** — Focus restoration, skip links, programmatic focus
- **Forced colors / Windows High Contrast** — `@media (forced-colors: active)` guidance
- **Reduced motion** — Documented but not applied systematically to each animated component
- **Screen reader test checklists** — VoiceOver (macOS/iOS), NVDA/JAWS (Windows)

5.3 Internationalisation

> **RTL Support — Complete Absence**
> Sitka has no right-to-left documentation, no bidirectionality guidance, and no CSS logical properties guidance. This is a hard blocker for Arabic, Hebrew, Farsi, and Urdu markets. Polaris and Lightning both have explicit RTL support. This is a one-time investment with high market reach impact.

- No locale-aware date/number/currency formatting guidance
- No text expansion allowance guidance (German text is ~35% longer than English)
- No pluralisation rules documentation

5.4 Density System

No compact / comfortable / spacious mode. Lightning DS has density variants throughout. This is a hard blocker for enterprise data-dense products — tables, forms, and lists all need density control. Recommended tokens:

- `--density-compact` — reduces padding by ~25%, line-height to 1.3
- `--density-default` — current Sitka baseline
- `--density-comfortable` — increases padding by ~30%, line-height to 1.8

5.5 Touch & Gesture Guidelines

Apple HIG is the gold standard here. Sitka's Interaction page covers hover and focus states but is missing:

- Gesture reference (tap, double-tap, long press, swipe, pinch, pan, rotate)
- Touch target minimum sizes — Apple requires 44×44pt; not documented in Sitka
- Haptic feedback taxonomy (impact, notification, selection feedback)
- Pointer type differentiation (mouse vs. touch vs. stylus)
- Scroll momentum and rubber-band behavior guidelines

6. Platform Gaps

PLATFORM	CURRENT STATUS	PRIORITY	EFFORT
Web (React)	Full coverage — Next.js + Vite libraries	—	—
iOS (SwiftUI)	Full code samples, dedicated library page	—	—
macOS (SwiftUI)	Full code samples, dedicated library page	—	—
Android (Jetpack Compose)	Not present	Medium	High
React Native	Not present — would bridge web + mobile	Medium	Medium
watchOS	Not present	Low	Medium
visionOS	Not present — yet Sitka's Liquid Glass aesthetic aligns perfectly	Medium	Medium
tvOS	Not present	Low	Low

> **visionOS Opportunity**
> Sitka's Liquid Glass language is already the closest any independent design system gets to Apple's visionOS aesthetic. Adding a visionOS foundations page — covering depth, materials, ornaments, and window chrome — would be a genuine first-mover differentiator requiring relatively little new design work given existing glass foundations.

Distribution Gap

Sitka is currently a documentation site only. There is no `npm install @sitka/ui`. Engineers must manually copy component code. The reference systems all have a distribution story:

- **Polaris** — `npm install @shopify/polaris`
- **Lightning** — `npm install @salesforce-ux/design-system`
- **shadcn** — CLI-based copy-paste (`npx shadcn@latest add button`)

A shadcn-style CLI is the lowest-effort distribution path: `npx sitka@latest add button` copies a component file into the user's project. This preserves the documentation-first model while giving engineers a repeatable install story.

7. Pattern Gaps

7.1 Missing High-Value Patterns

PATTERN	REFERENCE	PRIORITY
Multi-step Wizard / Stepper	Polaris, Lightning	High
Virtual / Infinite Scroll	Polaris, shadcn	High
Multi-select + Bulk Actions	Polaris, Lightning	High
Activity Feed / Timeline	Lightning (Activity Timeline)	Medium
Chat / Messaging UI	Industry standard	Medium
Search Results Page	Polaris	Medium
Inline Editing	Lightning	Medium
Pricing / Feature Comparison Table	Common SaaS pattern	Medium
Optimistic UI	Modern web best practice	Medium
Real-time / Live Data Indicators	—	Low
Offline / Sync States	—	Low
Onboarding Checklist	Common SaaS pattern	Low

7.2 AI Patterns — Extend the Head Start

Sitka's AI Agent Standards foundation is ahead of all four reference systems. These additions would extend that lead:

- **Streaming / typewriter text** — animation token (`--motion-stream-cursor`) + React component for progressive text reveal
- **AI chat conversation layout** — user message / assistant message / system message differentiation; threading; reaction affordances
- **AI-generated content labeling** — Badge variant (`variant="ai"`) for content provenance disclosure
- **Confidence / uncertainty indicator** — Visual treatment for probabilistic outputs
- **Prompt input component** — Multi-line input with submit-on-Enter, auto-height expansion, attachment affordance

8. Tooling & Governance Gaps

GAP	IMPACT	REFERENCE
No Storybook	Engineers cannot develop/test components in isolation	Polaris, Lightning, shadcn all use Storybook
No component lifecycle taxonomy	`badge: "New"` is the only status; no Alpha/Beta/Stable/Deprecated with migration guides	Polaris, Lightning
Figma library disabled	`FEATURES.figmaLibrary: false` — designers cannot use Sitka components in Figma	All four reference systems have Figma kits
No npm / CLI distribution	No repeatable install story for engineers	Polaris, Lightning, shadcn
No contributing guidelines	External contributions have no process	All open-source DSes
No browser support matrix	Engineers don't know what browsers are supported	Polaris, Lightning
No visual regression testing	No protection against unintended visual changes	Polaris (Chromatic), Lightning
No custom ESLint rules	Nothing prevents hardcoded hex colors or non-token values in consuming projects	Lightning has ESLint rules
No versioning / migration guides	No documented upgrade path between versions	Polaris, shadcn

9. Implementation Strategy

The gaps above are prioritised into four phases based on two criteria: **blocking potential** (do consumers hit this immediately?) and **effort-to-impact ratio**. P0 items should be resolved before any marketing push; P3 items are differentiators that can be built incrementally.

P0

Correctness Sprint

Estimated 2–3 weeks · Zero new design work

> **Goal: Nothing is broken or missing that already exists in the codebase**
> These are documentation-only tasks for code that already ships. The design system should not claim components that have no docs page.

#	TASK	TYPE	EFFORT
1	Document **Tabs** as standalone component page	COMPONENT	Low
2	Document **Select** as standalone component page	COMPONENT	Low
3	Promote **Checkbox, Radio, Switch** from Form Controls to standalone pages	COMPONENT	Low
4	Document **Command Palette** as a reusable component	COMPONENT	Low
5	Add **Textarea** component + docs	COMPONENT	Low
6	Move all semantic tokens into `tokens.json` — surface, text, border, accent, glass, z-index	TOKEN	Medium
7	Add motion tokens to `tokens.json` — duration scale + easing curves	TOKEN	Low
8	Add shadow tokens to `tokens.json` — card, lifted, sheet, overlay	TOKEN	Low

P1

Completeness Sprint

Estimated 4–6 weeks · Required for production-grade claim

> **Goal: No component that every DS has is missing from Sitka**
> After this phase, Sitka can claim full component parity with shadcn/ui and Polaris on common elements.

#	TASK	TYPE	EFFORT
9	**Alert / Banner** — persistent page-level feedback (info/success/warning/error)	COMPONENT	Low
10	**Accordion / Collapsible**	COMPONENT	Low
11	**Pagination**	COMPONENT	Low
12	**Spinner** (standalone inline loader)	COMPONENT	Low
13	**Divider / Separator**	COMPONENT	Low
14	**Label** (standalone form label)	COMPONENT	Low
15	**Toggle / Toggle Group**	COMPONENT	Low
16	**File Upload / Drop Zone**	COMPONENT	Medium
17	**Drawer** (horizontal slide-in panel, distinct from Bottom Sheet)	COMPONENT	Medium
18	**Sidebar** — document the docs-shell sidebar as a standalone component	COMPONENT	Medium
19	Add per-component **ARIA role specifications** to each existing component page	FOUNDATION	Medium
20	Add **keyboard navigation model** to each interactive component	FOUNDATION	Medium
21	**Multi-step Wizard** pattern	PATTERN	Medium
22	Enable **Figma library** (`FEATURES.figmaLibrary: true`) + publish Code Connect mappings	TOOLING	Medium

P2

Competitive Parity Sprint

Estimated 6–10 weeks · Required for enterprise adoption

Goal: Remove the blockers that prevent enterprise and international teams from adopting Sitka

#	TASK	TYPE	EFFORT
23	**Density system** — compact / comfortable modes via CSS custom property overrides	TOKEN	Medium
24	**RTL / Internationalisation** foundations page + CSS logical properties audit	FOUNDATION	Medium
25	**Layout primitives** — Box, Stack, Inline Stack components	COMPONENT	Medium
26	**Responsive grid system** — breakpoint tokens + 12-column grid documentation	FOUNDATION	Medium
27	**Navigation Menu** (mega-nav / multi-level)	COMPONENT	Medium
28	**Menubar** (desktop File/Edit/View style)	COMPONENT	Medium
29	**Data Grid** — advanced table with sorting, filtering, virtual scroll	COMPONENT	High
30	**Forced-colors / High Contrast** support for all components	FOUNDATION	Medium
31	**Storybook** integration — stories for all existing components	TOOLING	High
32	**Component lifecycle taxonomy** — Alpha / Beta / Stable / Deprecated badges + migration guides	TOOLING	Low
33	**CLI distribution** — `npx sitka@latest add <component>` install story	TOOLING	High
34	**Activity Feed / Timeline** pattern	PATTERN	Medium
35	**Multi-select + Bulk Actions** pattern	PATTERN	Medium

P3

Differentiation Sprint

Ongoing · Builds unique competitive moat

> **Goal: Make Sitka the reference design system for AI-native, glass-aesthetic, multi-platform products**

#	TASK	TYPE	EFFORT
36	**visionOS / Spatial Computing** foundations — depth, materials, ornaments, window chrome	PLATFORM	Medium
37	**AI Chat conversation layout** pattern	PATTERN	Medium
38	**Streaming / typewriter text** animation token + component	COMPONENT	Low
39	**Prompt input** component (multi-line, auto-expand, attachment-aware)	COMPONENT	Medium
40	**React Native** library — bridges web and mobile with shared tokens	PLATFORM	High
41	**Android / Jetpack Compose** library	PLATFORM	High
42	**Gesture reference** — comprehensive iOS/macOS/web gesture documentation	FOUNDATION	Medium
43	**Haptic feedback taxonomy** (impact, notification, selection)	FOUNDATION	Low
44	**Pricing / Feature comparison** pattern	PATTERN	Low
45	**Custom ESLint plugin** — lint against hardcoded hex colors and non-token values	TOOLING	Medium
46	**Input OTP / PIN** component	COMPONENT	Low
47	**Resizable panels** component	COMPONENT	Medium

10. Master Backlog Summary

P0 — CORRECTNESS

8 items · 2–3 weeks

- Document Tabs, Select, Checkbox, Radio, Switch, Command Palette
- Add Textarea component
- Move all semantic tokens into `tokens.json`
- Add motion / shadow / z-index tokens

P1 — COMPLETENESS

14 items · 4–6 weeks

- Alert/Banner, Accordion, Pagination, Spinner
- Divider, Label, Toggle, File Upload, Drawer, Sidebar
- Per-component ARIA + keyboard navigation specs
- Multi-step Wizard pattern
- Enable Figma library

P2 — PARITY

13 items · 6–10 weeks

- Density system, RTL/i18n, layout primitives, grid
- Navigation Menu, Menubar, Data Grid
- Forced-colors support
- Storybook, CLI distribution, lifecycle taxonomy
- Activity Feed, Multi-select patterns

P3 — DIFFERENTIATION

12 items · Ongoing

- visionOS foundations
- AI chat layout + streaming text + prompt input
- React Native + Android libraries
- Gesture reference + haptics
- ESLint plugin, OTP input, Resizable panels

Effort vs Impact Matrix

PRIORITY	ITEMS	DESIGN WORK	ENGINEERING WORK	MARKET IMPACT
P0	8	None	Low	High
P1	14	Low–Medium	Medium	High
P2	13	Medium	High	Medium–High
P3	12	Medium	High	Differentiating

Recommended Starting Point
Start with P0. All eight items can be completed in a single sprint with no new design decisions — they are documentation tasks for code that already exists. Completing P0 closes the most embarrassing gaps (undocumented existing components, incomplete token file) before any external sharing or marketing activity. P1 follows immediately after.